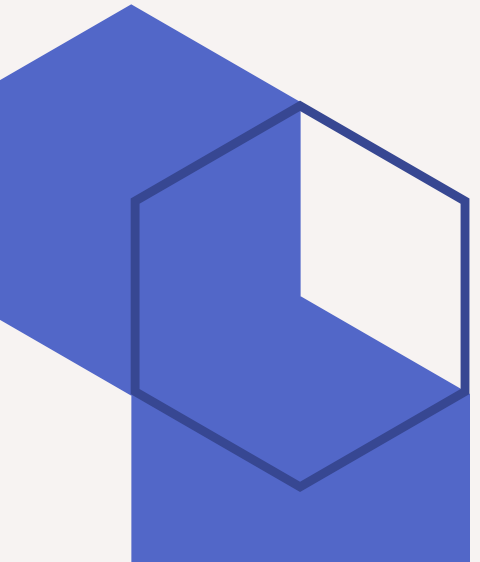




Analyzing Contributing Factors to MTR Fare Pricing

By:
Winiera Sutanto
(3035964671)



East Rail Crossing the Harbour | Looking only at MTR fares: "Cheap in the East, Expensive in the West" is a misplaced issue

Written by: Tang Wenshi

publishing: 2022-05-06 07:00 renew: 2025-02-17 20:14



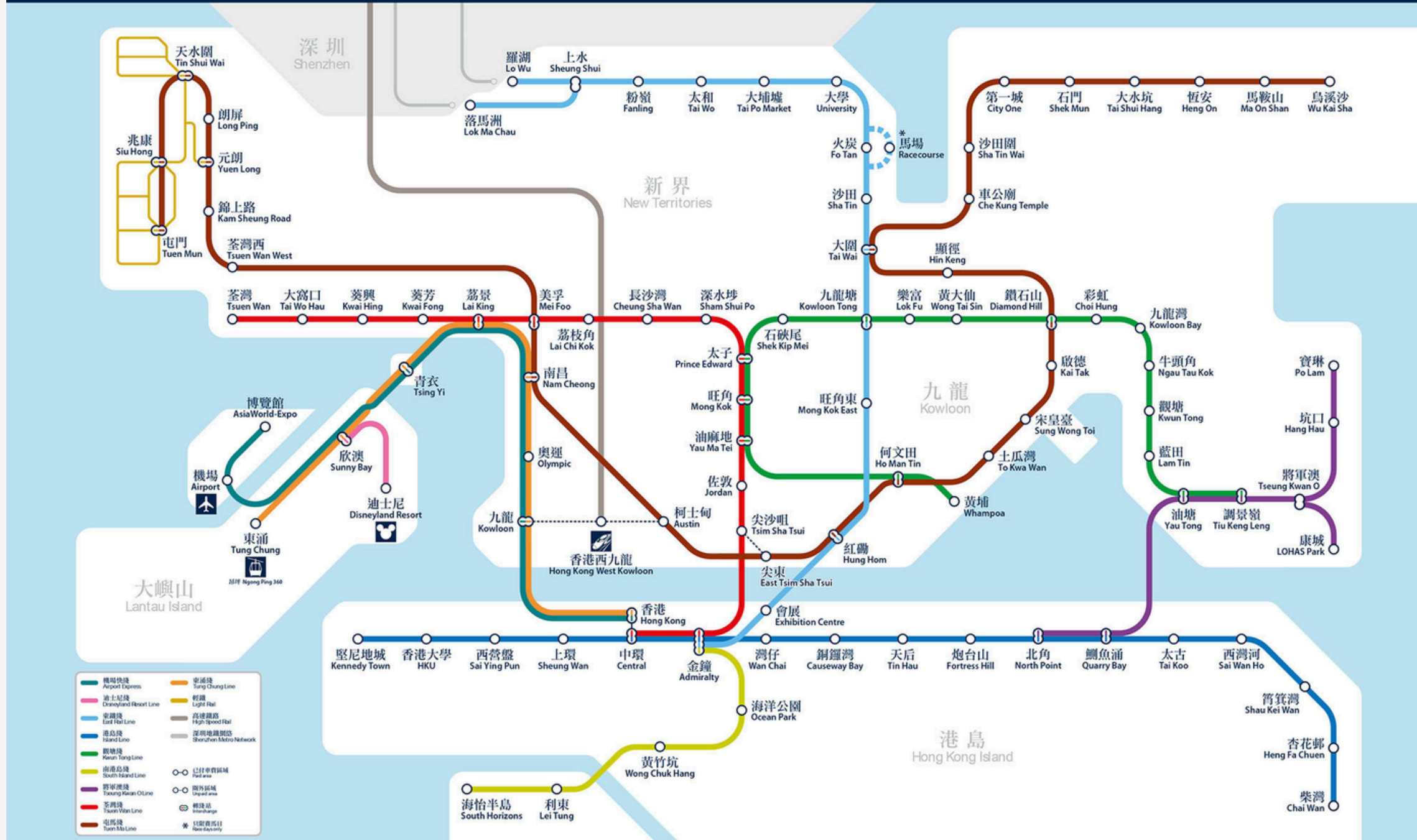
The opening of the East Rail Cross-Harbour section has led to a rise in prices in the east and west. Tin Pei-chun criticised the government for being irresponsible in slamming the MTR.

Translated to English

Major Questions

- What factors affect the pricing of MTR fares?
- Are certain districts more expensive than others? Why?

港鐵路綫圖 MTR system map



Possible Factors

- Number of stops?
- Distance of line?
- Inter-district travel?
- Population density?

Datasets

MTR routes, fares and barrier-free facilities

MTR Lines (except Airport Express & Light Rail) Fares

- Starting Station
- Destination Station
- Octopus Fare

Hong Kong MTR / Subway Network

All MTR stations and lines. Including geographical coordinates and distances

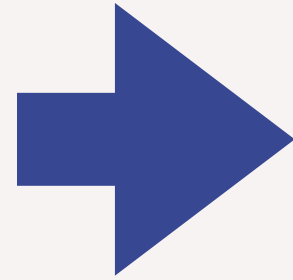
- Distance (in km)
between stations

Population and Household Statistics Analysed by District Council District

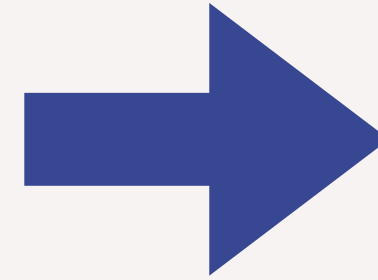
- Median Income per district
- Population Density per district

Workflow

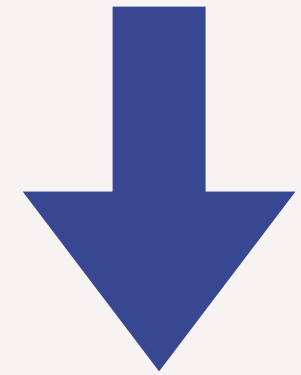
Obtain MTR dataset from
government website
(Routes and Fares)



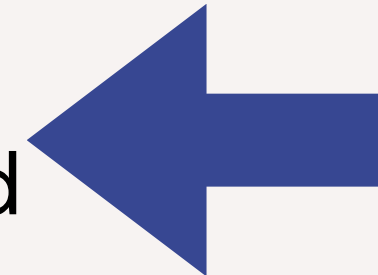
Scraped Wikipedia to
obtain station
coordinates



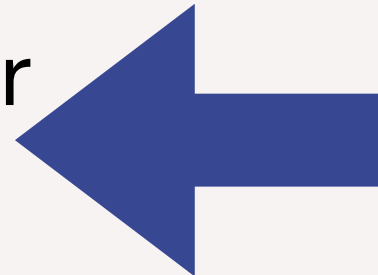
Used coordinates to find
distance between stations
and respective districts



Conducted EDA and
cleaned data (Removed
airport line, Lo Wu, and
Lok Ma Chau,
Racecourse)



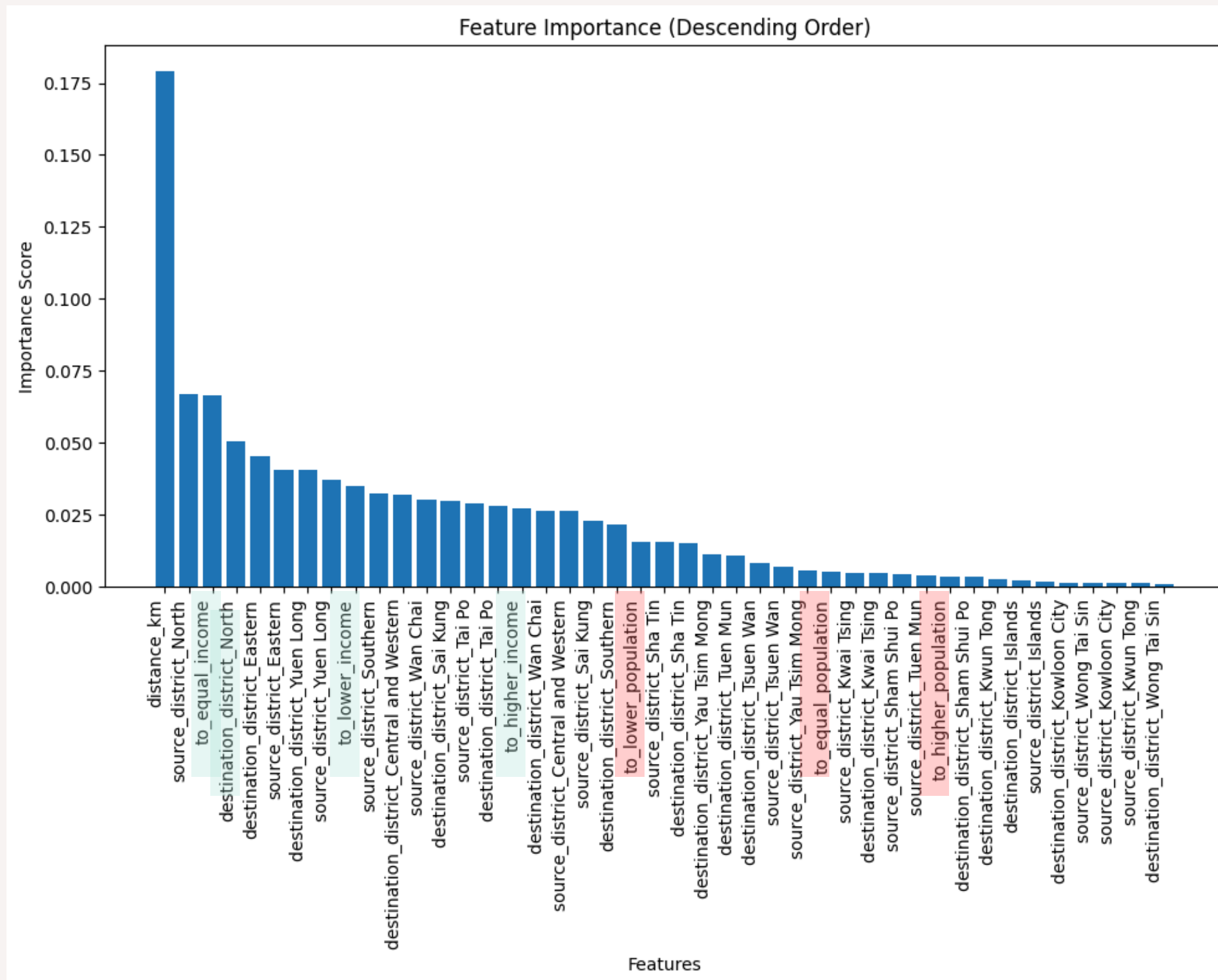
Added socioeconomic
data (District level
population density and
income median)



Use modeling to answer
the initial questions

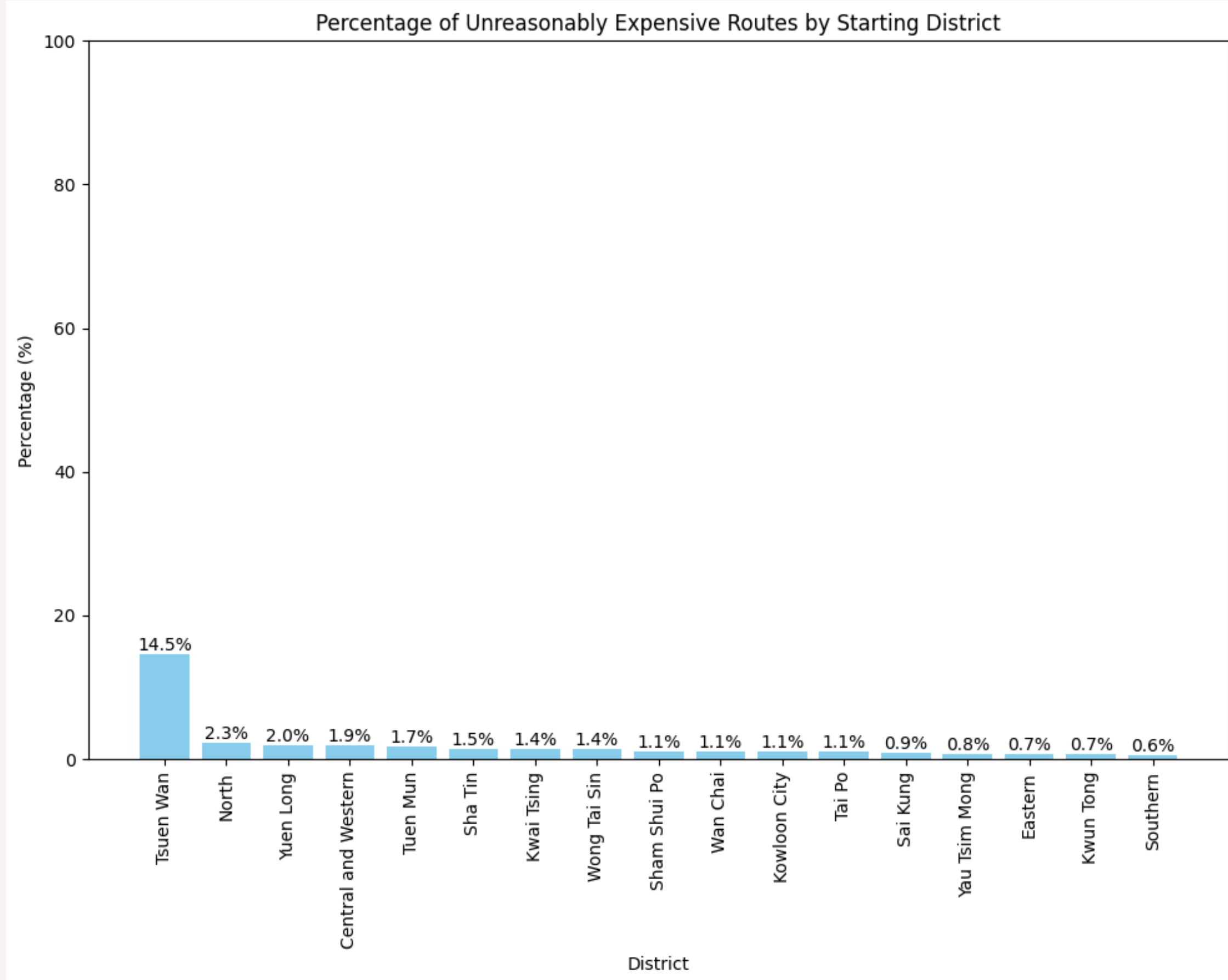
Which factors are the most important?

Chosen Model: XGBoost



- All models agree that distance is the most important factor
- R squared: 0.92
- Median Income is more important compared to population density
- Traveling from/to North and Eastern districts is most important

Comparison of Different Districts



- Look at routes in a district (Aggregating might not paint the full picture)
- Unreasonably expensive: 1.5 IQR above upper quartile
- 153/7832 routes considered unreasonably expensive
- Tseun Wan district has the highest percentage

Why is Tsuen Wan District so expensive?

Sunny Bay 欣澳

✕ MTR rapid transit station

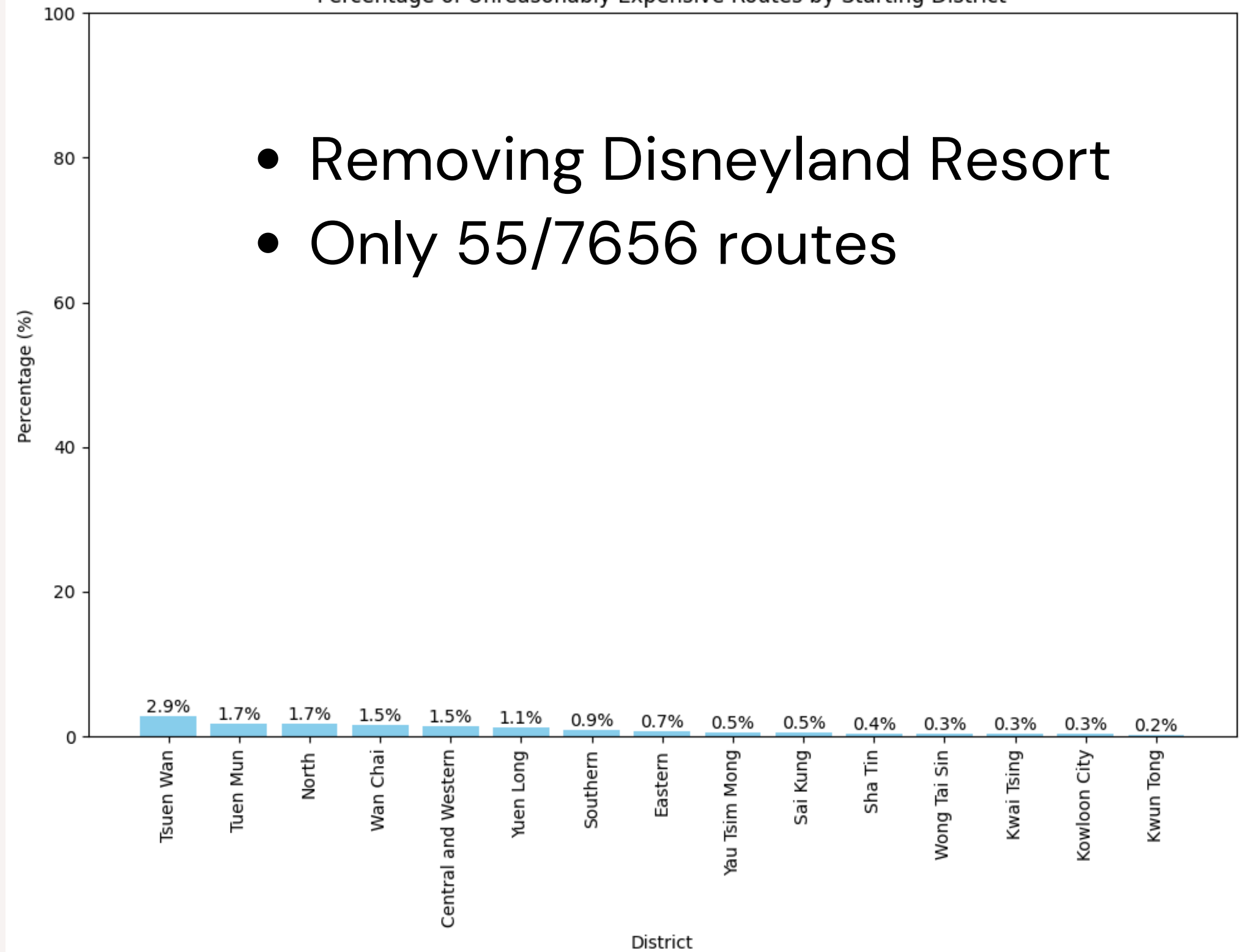
Location Sunny Bay Road, Yam O, Lantau Island
Tsuen Wan District, Hong Kong

Disneyland Resort 迪士尼

✕ MTR rapid transit station

Location Fantasy Road, Penny's Bay, Lantau Island
Tsuen Wan District, Hong Kong

Percentage of Unreasonably Expensive Routes by Starting District



- Removing Disneyland Resort
- Only 55/7656 routes

Conclusion

- Distance is the most important factor
- Only around 0–2% of MTR routes in a district can be considered unreasonably expensive

Limitations

- Does not account for subsidy schemes
- Some routes have alternative public transportation
- Factors like passenger capacity and train frequency could also affect pricing